

REAL ESTATE ANALYTICS | MACHINE LEARNING | DATA SCIENCE

Explainable Real Estate Analytics: A Comparative Analysis of Machine Learning Frameworks for House Price Prediction

BATCH – 2

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PROJECT REVIEW – 2

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AGENDA

What This Review Covers

01

Methodology

Refined end-to-end pipeline executed since Review-1

02

System Design

Layered architecture of the analytics pipeline

03

Modern Tools & Technologies

Stack used to build and evaluate the system

04

Implementation

Modules completed and work currently in progress

05

Results

Snapshot of model outputs generated so far

06

Performance Evaluation

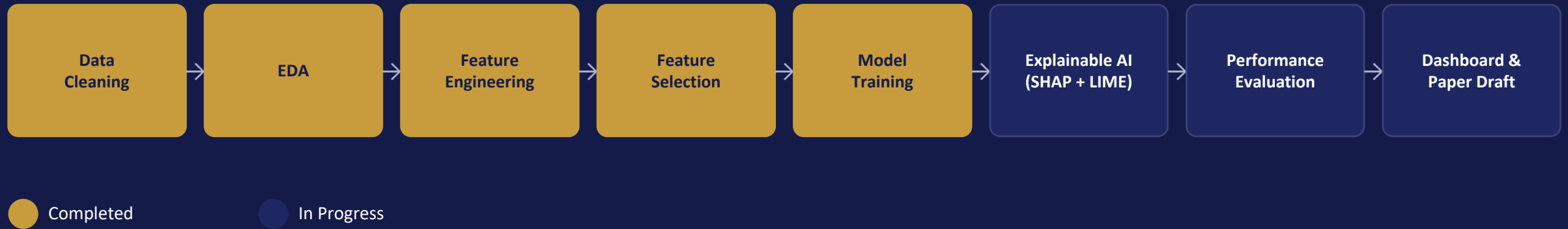
Framework used to compare models fairly

07

Research Paper Draft Preparation

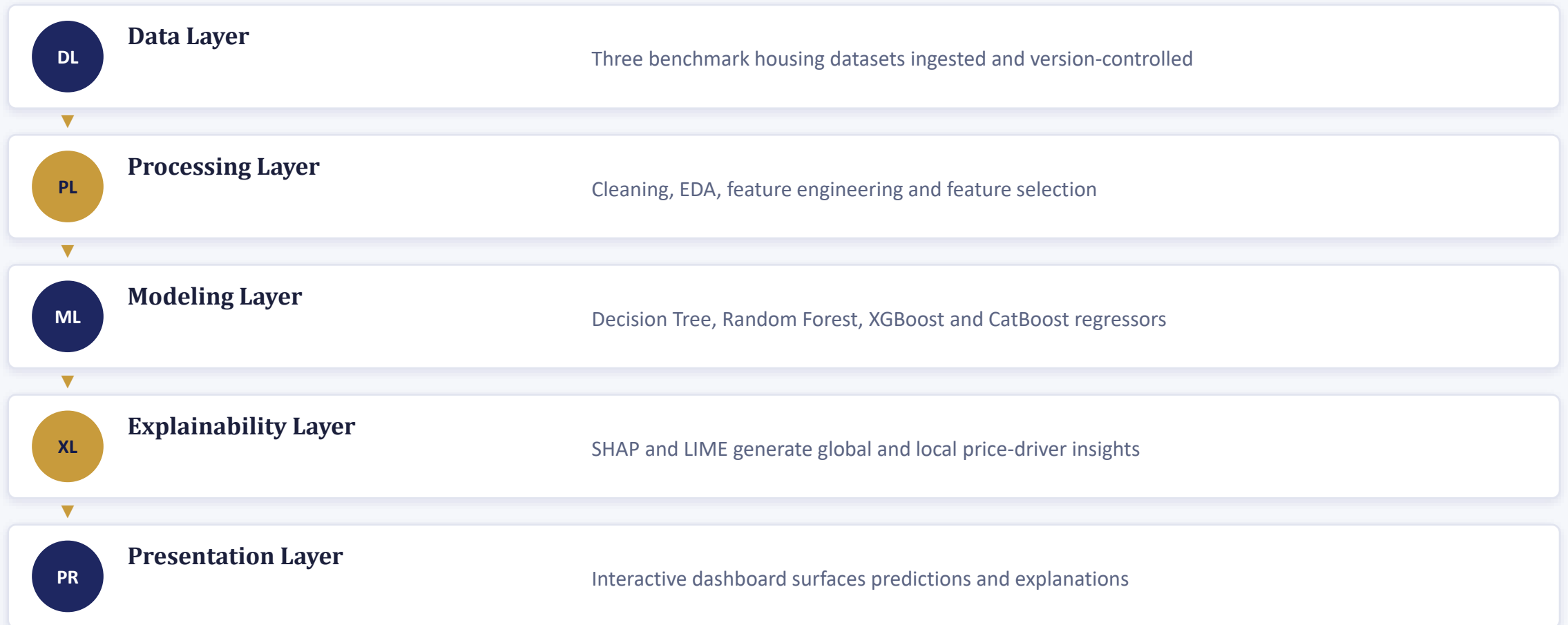
Status of the manuscript in progress

Refined Data-to-Insight Pipeline



Benchmark data across three housing datasets has been cleaned, explored, engineered and used to train baseline models. Boosted-tree models, explainability integration, comparative evaluation and the dashboard/manuscript are the current focus.

End-to-End System Architecture



Technology Stack



Language & Environment

Python 3, Jupyter Notebook, VS Code



Data Handling

Pandas, NumPy



Machine Learning

Scikit-learn, XGBoost, CatBoost



Explainable AI

SHAP, LIME



Visualization

Matplotlib, Seaborn, Plotly



Deployment & Collaboration

Streamlit, Git & GitHub

Implementation Progress

COMPLETED

- Data ingestion and cleaning scripts for all three benchmark datasets
- Exploratory Data Analysis notebooks (univariate, bivariate, multivariate)
- Feature engineering and correlation-based feature selection pipeline
- Train-test split with consistent preprocessing (encoding, scaling)
- Baseline Decision Tree and Random Forest models trained

IN PROGRESS

- Hyperparameter tuning for XGBoost and CatBoost regressors
- SHAP and LIME integration for model explainability
- Comparative performance evaluation across all three datasets
- Interactive Streamlit dashboard prototype
- Research paper draft (methodology and results sections)

RESULTS

Model Output Snapshot

Final values to be filled in once evaluation across all three benchmark datasets is complete.

Dataset	Model	MAE	RMSE	R ²	MAPE
Dataset 1	Decision Tree	162,026.47	324,510.80	0.2921	28.94%
	Random Forest	129,897.10	313,921.89	0.3376	27.03%
	XGBoost	132,034.61	335,203.24	0.2447	27.79%
	CatBoost	136,290.03	244,988.48	0.5966	28.48%
Dataset 2	Decision Tree	343,749.57	417,633.54	-1.2419	127.87%
	Random Forest	254,841.22	293,554.62	-0.1077	105.78%
	XGBoost	250,915.56	291,074.90	-0.0890	104.61%
	CatBoost	246,488.78	284,742.56	-0.0422	105.13%
Dataset 3	Decision Tree	135,300.96	277,616.02	0.5261	24.76%
	Random Forest	99,668.49	198,831.54	0.7569	18.39%
	XGBoost	99,891.48	201,090.47	0.7514	18.43%
	CatBoost	122,075.08	229,419.41	0.6764	23.79%

Navigation

- Dashboard
- Dataset Explorer
- Data Visualization
- Feature Analysis
- Cross-Dataset Comparison
- Model Performance
- Explainable AI
- Predict House Price
- ML Concepts

3 Real-World Datasets • 5 ML Models • 15 Experiments

Dataset-Level Model Comparison

Select Benchmark Dataset

Dataset 1

Best Model

Stacking Regressor

R²

0.6505

RMSE

228,039

MAPE

28.58%

Performance Metrics — Dataset 1

Model	R2 Score	MAE	MSE	RMSE	MAPE	CV R2 Mean	CV R2 Std
Stacking Regressor	0.6505	133832.06	52001581718.54	228038.55	28.58	0.4215	0.2206
Linear Regression	0.6378	138384.89	53877441979.78	232115.15	29.85	0.4464	0.232
CatBoost	0.5966	136290.03	60019356831.27	244988.48	28.48	0.385	0.2069
Random Forest	0.3376	129897.1	98546951446.04	313921.89	27.03	0.3191	0.227
Decision Tree	0.2921	162026.47	105307259515.68	324510.8	28.94	-0.7941	1.8525
XGBoost	0.2447	132034.61	112361210331.16	335203.24	27.79	0.131	0.3398

R² Score Comparison

Model R² Scores — Dataset 1



Real Estate Analytics

DAV Machine Learning Project

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3 Real-World Datasets • 5 ML Models • 15 Experiments

1. Ensemble Learning

The **Stacking Regressor** combines predictions from multiple base learners instead of depending on only one algorithm.

The ensemble uses:

- Linear Regression
- Random Forest
- XGBoost

A Ridge regression model is used as the final meta-learner.

Stacking Performance

Dataset	R2 Score	MAE	RMSE	MAPE	CV R2 Mean	CV R2 Std
Dataset 1	0.6505	133832.06	228038.55	28.58	0.4215	0.2206
Dataset 2	-0.0026	242914.83	279282.13	103.97	-0.0102	0.0097
Dataset 3	0.7651	97185.9	195448.66	17.75	0.7482	0.0329

Strongest Overall Experiment

Model

Dataset

Test R²

CV R²

Stacking Regressor

Dataset 3

0.7651

0.7482

Stacking Regressor on Dataset 3 achieved the strongest overall test R² of 0.7651.

Stacking vs Individual Models

Select Dataset

How Models Will Be Compared

1

Error-Magnitude Metrics

MAE, MSE and RMSE quantify average prediction error in price units.

2

Goodness-of-Fit

R^2 Score measures how well each model explains price variance.

3

Scale-Independent Error

MAPE enables fair comparison across datasets of different price ranges.

4

Cross-Validation

K-fold validation checks that results generalize beyond a single split.

5

Cross-Dataset Consistency

Every model is benchmarked on all three datasets, not just one.

6

Explainability Consistency

SHAP feature-importance rankings are compared across models.

Manuscript Status

Abstract & Introduction

Drafted

Literature Review

In Progress

Methodology

Drafted

Results & Discussion

Pending Final Results

Conclusion & Future Work

Pending

NEXT MILESTONES

- Finalize model results across all datasets
- Complete SHAP/LIME analysis write-up
- Internal review of full draft
- Submit to target conference / journal



Thank You

Questions & Feedback Welcome